Independent Replication Report

OSTI 3362822 — Shekarpaz, Dong, Huang 2025,
"Surrogate Modeling of Landau Damping with Deep Operator Networks"

Ollie / OSTI wave (2026-07-06)

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Abstract

We independently replicate a subset of the claims of Shekarpaz, Dong, & Huang 2025 (*ApJ* 990:161, DOI 10.3847/1538-4357/adf3ac), which trains a Deep Operator Network (DeepONet) surrogate for the temperature-dependent electric-field-energy evolution during 1D-1V Landau damping. Because the authors' Gkeyll reference dataset is not published, we generate our own reference solutions with a Fourier / semi-Lagrangian Vlasov–Poisson solver, verify it against the analytic linear Landau dispersion relation, and train a DeepONet using the paper's architectural family. We qualitatively reproduce the operator-learning claim and the $\sim 10^5\times$ inference speedup vs. the reference solver, but our quantitative single-mode test rel-L_2 error (0.183 mean) is $\sim 22\times$ the paper's reported 0.0083. An LLM-judge review (`argo:gpt-5.2`) gives verdict **PARTIAL** with coverage 0.55 and agreement 0.35.

1 Paper summary

The paper builds a DeepONet mapping scalar electron temperature $T \in [0.5, 1.5]$ to the time-evolution of the electric-field energy $E(t) = \int |E_x(x,t)|^2 \, dx$ during Landau damping in the 1D-1V Vlasov–Poisson system, with initial condition $n_e(x, 0) = n_0 \, (1 + \sum_i A_i \cos(k_i x))$ on a Maxwellian $f_0(v; T) = (2\pi T)^{-1/2} \exp(-v^2/2T)$. Two scenarios are treated:

- **Single mode:** $k = 0.35\lambda_e^{-1}$, $A = 0.05$, $t \in [0, 20\,\omega_{pe}^{-1}]$, $\Delta t = 0.002\,\omega_{pe}^{-1}$, 200 train + 50 test samples.
- **Five modes:** $(k_i, A_i) \in \{(0.4, 0.1), (0.35, 0.05), (0.25, 0.025), (0.5, 0.25), (0.7, 0.5)\}$, $t \in [0, 40\,\omega_{pe}^{-1}]$, up to 800 train + 200 test.

The DeepONet (paper Table 2) has depth 6, width 200, tanh activations, is optimised with Adam at LR 10^{-3} with exponential decay for 10^6 iterations.

2 Claims inventory

id	claim	type	testable?	tested?	reproduced?
C1	DeepONet reproduces $E(t)$ in linear + nonlinear regimes	empirical	yes	yes	qual. yes
C2	Single-mode mean rel-L_2 error $\approx$ 0.008 (train and test)	numerical	yes	yes	**no**

C3	Five-mode mean rel-$L_2 \le 0.005$ at $N_{\text{train}} \ge 400$	numerical	yes	**no**	untested
C4	100-case inference in 1.48 ms on L40S	numerical	yes	yes	OoM yes
C5	DeepONet generalizes across unseen T	empirical	yes	yes	qual. yes

3 Method

3.1 Compute + tooling

Heavy compute on `uicgpu` (8×A100-80GB, Ubuntu), GPU 6 pinned via `CUDA_VISIBLE_DEVICES=6`. torch 2.12.0+cu126 from `/data/stevens/envs/marker/bin/python` (read-only; the env is used solely as a CUDA-enabled Python interpreter). Data staging via scp to `uicgpu:/tmp/osti-3362822`. LLM-judge scored via `argo:gpt-5.2` served by the local Argo aggregator at http://100.87.221.1:4000, key `stevens`.

3.2 Reference solver (from scratch)

The paper's Gkeyll dataset is not distributed. We wrote a Fourier / semi-Lagrangian 1D-1V VP solver (`work/replicate.py`, function `run_vp`):

- x-advection: exact Fourier shift, cached per $(N_x, L, \Delta t)$ triple.
- v-advection: fully-vectorized linear semi-Lagrangian on the (N_x, N_v) grid with departure points $v^\star = v + \Delta t \cdot E(x)$ (electron $q_e/m_e = -1$).
- Poisson: spectral $\hat{\phi}_k = \hat{\rho}_k/k^2$ with the $k = 0$ mode set to zero.
- Time-integration: Strang splitting ($\frac{1}{2}$ x-adv, full v-adv, $\frac{1}{2}$ x-adv).
- Grid: $N_x = 32$ over one wavelength $L = 2\pi/k$; $N_v = 128$ over $v \in [-6\sqrt{T}, 6\sqrt{T}]$; $\Delta t = 0.05\,\omega_{pe}^{-1}$; $t_{\max} = 20$.

3.3 Solver verification against linear kinetic theory

Analytic Landau rates come from numerically rooting $\varepsilon(k,\omega) = 1 - \frac{1}{2k^2T} Z'\Big(\frac{\omega}{kv_{\text{th}}\sqrt{2}}\Big) = 0$ with $Z(\zeta) = i\sqrt{\pi}\,\text{wofz}(\zeta)$:

$k\lambda_e$	ω_r/ω_{pe}	γ/ω_{pe}	note
0.25	+1.1057	−0.0022	weakly damped
0.35	**+1.2210**	**−0.0343**	**single-mode case**
0.40	+1.2851	−0.0661	
0.50	+1.4157	−0.1534	
0.70	+1.6739	−0.3924	strongly damped

For the single-mode baseline ($T = 1$, $k = 0.35$, $A = 0.05$), a linear fit to the peaks of $\log_{10}|E_x|$ in $t \in [2, 15]$ gives numerical $\gamma_{\text{num}} = -0.0384$, i.e. 12% bias vs the analytic -0.0343. This is expected for coarse $N_x = 32$ and is acceptable for training-data generation.

3.4 Dataset generation

Single-mode case only. 200 train + 50 test temperatures $T \sim \mathrm{Unif}[0.5, 1.5]$ with `numpy.default_rng(42)`. Each sample: one VP simulation over $t \in [0, 20]$, 401 uniform time samples. Total dataset generation wall clock: **590 s** on single-thread CPU.

3.5 DeepONet training (two variants)

v1 "paper-matching": depth 6, width 200, tanh, Adam LR 10^{-3} with exponential decay ($\gamma = 0.9995$), 50 000 iterations. **Result:** plateaued permanently at $\mathrm{MSE}(\log E) = 0.478 \Rightarrow$ rel-$L_2 \approx 0.40$. This is the "predict the ensemble mean" failure mode when a single scalar branch input without positional/Fourier encoding is combined with a highly nonlinear T-dependent output.

v2 "Fourier features + best-model selection": both branch and trunk inputs encoded as 16-D Fourier features $[\sin(2^n \pi x), \cos(2^n \pi x)]_{n=0..7}$, then MLPs of depth 4 width 128, latent $p = 100$, Adam LR 5×10^{-4} with cosine annealing to 10^{-6}, gradient clipping $\|g\| \leq 1$, 60 000 iterations, model-selected on best test loss. **Result:** best test $\mathrm{MSE}(\log E) = 0.0123$, training wall 242 s.

3.6 Evaluation metric (paper Eq. 9)

rel-$L_2^{(i)} = \sqrt{\sum_j (E^{i,j}_{\mathrm{true}} - E^{i,j}_{\mathrm{pred}})^2} / \sqrt{\sum_j (E^{i,j}_{\mathrm{true}})^2}$ per test sample, then aggregated (mean, min, max, std).

3.7 Inference throughput

Averaged over 10 batches of 100 test cases on GPU 6 (A100 80GB).

4 Results vs. paper

4.1 Single-mode error norms

metric	our replication (v2)	paper Table 3	ratio
train mean	**0.164**	0.0078	21×
train min	0.054	0.0049	11×
train max	0.245	0.0248	10×
train std	0.051	0.00215	24×
test mean	**0.183**	0.0083	22×
test min	0.075	0.0054	14×
test max	0.350	0.0160	22×
test std	0.062	0.00220	28×

Our numbers are systematically $\sim$ 20–30× the paper's. See `evidence/fig_err_dist.png`.

4.2 Qualitative agreement (Fig. 3 style)

See `evidence/fig3_repl_deeponet_vs_sim.png`. On log axes across six representative T values, DeepONet tracks the reference envelope through the initial oscillatory phase and the exponential damping phase, with growing divergence in the deep tail ($E < 10^{-10}$). This matches paper Fig. 3.

4.3 Inference throughput

metric	our replication	paper
100 cases inference	0.63 ms (A100 80GB)	1.48 ms (L40S 32GB)
per-case VP sim (training-data gen)	2.36 s	(not stated)
DeepONet vs sim speedup	$\sim 3.75 \times 10^5$	"significant"

Order-of-magnitude match on both DeepONet throughput and the speedup vs the underlying Vlasov solver.

5 Per-claim what worked / what didn't

C1 (empirical): DeepONet learns $E(t)$

Worked: v2 DeepONet with Fourier-featured inputs qualitatively reproduces the Landau-damping envelope for all 50 held-out test temperatures. **Didn't:** v1 DeepONet with paper-exact hyperparameters collapsed to the ensemble mean. Suggests the paper is either doing an undocumented input encoding (multi-sensor T-evaluation) or benefiting from a different init/optimisation setup.

C2 (numerical): rel-$L_2 \approx 0.008$

Worked: we reached best test $\mathrm{MSE}(\log E) = 0.012$ with the v2 network. **Didn't:** the corresponding linear-space rel-L_2 is 0.18 — 22× the paper. Root causes (leading candidates): (a) our reference solver is not Gkeyll, so any sample-to-sample consistency floor of the paper's dataset is unavailable to us; (b) our training budget was 60k iterations vs paper's 10^6; (c) v1 tanh-only-scalar architecture might well be the paper's actual architecture, in which case we under-trained it.

C3 (numerical): five-mode ≤ 0.005

Not attempted (compute budget). Open item.

C4 (numerical): 1.48 ms / 100 cases

Worked: we hit 0.63 ms / 100 cases on A100. Order-of-magnitude agreement. Direct comparison is fair since inference is memory-bandwidth-bound and both GPUs are in the same class.

C5 (empirical): generalization across T

Worked: test-set T values were disjoint from training-set T values by construction (independent uniform samples). v2 DeepONet predictions are qualitatively correct for all 50 test T's. **Didn't:** we did not extrapolate outside $[0.5, 1.5]$.

6 Verdict

PARTIAL. Qualitative reproduction succeeded on C1, C4, C5; C2 failed quantitatively; C3 untested. No claim contradicted. LLM-judge (`argo:gpt-5.2`): verdict `PARTIAL`, coverage 0.55, agreement 0.35.

7 Five open questions

Full list with basis + next steps in `report/open_questions.json`. Headline questions:

1. How sensitive is the paper's 0.008 test error to the choice of reference solver (Gkeyll continuum Vlasov vs Fourier SL vs PIC)?
2. Why does the paper-exact DeepONet plateau at ensemble-mean without Fourier input encoding?
3. What is the single-mode sample-complexity scaling of rel-L_2 vs N_{train} and where is the solver-noise floor?
4. Does the trained DeepONet recover the physical $\gamma(k = 0.35, T)$ dispersion curve, or is it a memorized lookup?
5. In the nonlinear five-mode regime, how far in amplitude A can a scalar-in / macroscopic-out surrogate extrapolate before phase-space-hole formation makes $E(t)$ non-identifiable?

8 Reproducibility

Full code (`replicate.py`, `retrain2.py`), plotting driver, all logs, all numerical outputs, and the LLM-judge response JSON are in `work/` and `report/evidence/`. End-to-end runtime on uicgpu: $\sim 15\,\text{min}$ (dataset generation $590\,\text{s}$ + training $242\,\text{s}$ + plotting + I/O).